CHAPTER 6

HALOALKANES AND HALOARENES

CHAPTER ESSENCE AT A GLANCE:
Mechanisms of nucleophilic substitution (SN1 and SN2) in haloalkanes, elimination reactions, and nucleophilic aromatic substitution in haloarenes.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Nucleophilic Substitution Mechanisms:** Primary halides favour bimolecular SN2 substitution proceeding through a concerted transition state with complete Walden inversion, whereas tertiary halides favour unimolecular SN1 substitution via planar carbocation intermediates yielding racemisation.

• **Saytzeff Elimination & Regioselectivity:** Dehydrohalogenation of alkyl halides by strong bases preferentially yields the most stable, most highly substituted alkene bearing the maximum number of alkyl substituents around the double bond.

• **Inertness of Haloarenes to Substitution:** Haloarenes resist nucleophilic displacement due to resonance-induced partial double bond character of the carbon-halogen bond, sp2 hybridization of the aromatic ring carbon, instability of the phenyl cation, and electron repulsion by the aromatic pi cloud.

• **Electrophilic Aromatic Substitution:** The halogen atom acts as an ortho-para directing group due to positive resonance donation, while simultaneously acting as a ring-deactivating substituent because of strong inductive electron withdrawal.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\text{Rate} = k[\text{R-X}] \quad (\text{S}_\text{N}1:\ 3^\circ > 2^\circ > 1^\circ,\ \text{Racemisation})$$

Kinetics and reactivity order of unimolecular nucleophilic substitution

$$\text{Rate} = k[\text{R-X}][\text{Nu}^-] \quad (\text{S}_\text{N}2:\ 1^\circ > 2^\circ > 3^\circ,\ \text{Walden Inversion})$$

Kinetics and transition state of bimolecular nucleophilic substitution

$$\text{R-X} + \text{NaI} \xrightarrow{\text{dry acetone}} \text{R-I} + \text{NaX}\downarrow \quad (\text{Finkelstein})$$

Finkelstein halogen exchange reaction driven by sodium halide precipitation

$$\text{R-Br} + \text{AgF} \rightarrow \text{R-F} + \text{AgBr}\downarrow \quad (\text{Swarts reaction})$$

Swarts synthesis of alkyl fluorides using metallic fluorides

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** S N1 leads to RACEMISATION (50% retention, 50% inversion) due to planar sp^2 carbocation attack from both faces.

• **Alert:** [Valuation Rubric] Classify substrate as 1°, 2°, or 3° alkyl halide .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain the SN1 mechanism for nucleophilic substitution of tertiary alkyl halides with energy profile and stereochemistry. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Step 1 (Slow or RDS)]** Ionization of 3° alkyl halide: $(CH_3)_3C$-Br --> $(CH_3)_3C^+$ (planar carbocation) + Br^-. Rate = k [$(CH_3)_3C$-Br]. [1.5 Marks]

• **[Step 2: Step 2 (Fast)]** Attack of nucleophile: $(CH_3)_3C^+ + OH^-$ --> $(CH_3)_3C$-OH. Planar intermediate allows attack from both sides, yielding racemisation (50% retention, 50% inversion). Reactivity: 3° > 2° > 1°. [2.5 Marks]

• **Key Exam Result:** SN1: 2-step unimolecular, carbocation intermediate, 3° > 2° > 1°, racemisation

2. Explain the SN2 mechanism for nucleophilic substitution of primary alkyl halides with inversion of configuration (Walden inversion). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Concerted Mechanism]** Nucleophile OH^- attacks primary alkyl halide from back side opposite to leaving group: $OH^- + CH_3$-Cl --> [HO···CH_3···Cl]‡ (pentacoordinate transition state) --> HO-$CH_3 + Cl^-$. [2.5 Marks]

• **[Step 2: Kinetics & Stereochemistry]** Rate = k [CH_3Cl] [OH^-] (Bimolecular 2nd order). Complete inversion of configuration (Walden Inversion). Reactivity order: CH_3X > 1° > 2° > 3° (governed by steric hindrance). [1.5 Marks]

• **Key Exam Result:** SN2: 1-step bimolecular, transition state, 1° > 2° > 3°, Walden Inversion

3. Compare SN1 and SN2 reaction mechanisms on the basis of: kinetics, order of reactivity, solvent, and stereochemistry. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Comparison Table]** 1. Kinetics: SN1 is 1st order (Rate = k[RX]); SN2 is 2nd order (Rate = k[RX][Nu^-]). 2. Intermediates: SN1 forms carbocation; SN2 forms transition state. 3. Reactivity order: SN1: 3° > 2° > 1°; SN2: 1° > 2° > 3°. 4. Stereochemistry: SN1 gives racemisation; SN2 gives complete inversion. [3 Marks]

• **Key Exam Result:** SN1 (3°>2°>1°, racemic) vs SN2 (1°>2°>3°, inversion)

4. Explain Saytzeff's (Zaitsev's) rule for dehydrohalogenation of alkyl halides with an example. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Statement]** In dehydrohalogenation (β-elimination) of alkyl halides, the preferred product is the alkene having the greatest number of alkyl groups attached to the double-bonded carbon atoms (more substituted/stable alkene). [1 Mark]

• **[Step 2: Example]** 2-Bromopentane + alc. KOH --> pent-2-ene (81%, major) + pent-1-ene (19%, minor). [1 Mark]

• **Key Exam Result:** Saytzeff's Rule: More substituted alkene is major product

5. Explain why haloarenes are extremely less reactive towards nucleophilic substitution compared to haloalkanes (four reasons). [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Four Reasons]** 1. Resonance effect: C-Cl bond acquires partial double bond character, making it shorter and stronger. 2. Difference in hybridization: Aromatic carbon is sp^2 (more electronegative) holding C-Cl tighter than sp^3 carbon in alkyl halide. 3. Instability of phenyl cation formed by self-ionization. 4. Electrostatic repulsion between incoming electron-rich nucleophile and benzene π-electron cloud. [3 Marks]

• **Key Exam Result:** Partial double bond character, sp^2 carbon, phenyl cation instability, π repulsion

6. Explain the preparation and synthetic use of Grignard reagent (RMgX). Why should it be prepared under anhydrous conditions? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Preparation]** R-X + Mg --> [dry ether] R-Mg-X (Grignard reagent). [1 Mark]

• **[Step 2: Anhydrous Condition]** Highly reactive polar organometallic bond ($R^{\delta-}$-$Mg^{\delta+}X$). Must be strictly anhydrous because it reacts violently with any active hydrogen or moisture to give corresponding alkane: R-Mg-X + $H_2O \rightarrow$ R-H + Mg(OH)X. [2 Marks]

• **Key Exam Result:** RMgX + H_2O --> R-H + Mg(OH)X; Water destroys Grignard reagent

7. Predict the order of reactivity towards SN2 displacement: 1-bromobutane, 2-bromobutane, 2-bromo-2-methylpropane. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Steric Hindrance Principle]** SN2 reactivity is controlled by steric hindrance at reaction center (1° > 2° > 3°). [1 Mark]

• **[Step 2: Order Evaluation]** 1-bromobutane (1°, unbranched) > 1-bromo-2-methylpropane (1°, β-branched) > 2-bromobutane (2°) > 2-bromo-2-methylpropane (3°). [1 Mark]

• **Key Exam Result:** 1-bromobutane > 2-bromobutane > 2-bromo-2-methylpropane